

DOCKERFILE

A Dockerfile is a text file that contains instructions used to automatically build a Docker image. It defines the base image, application files, dependencies, commands, and configurations required to create and run a container.

- A script used to build a Docker image automatically.
- Contains a set of instructions written in order.
- Instructions are usually written in uppercase.
- Order of instructions matters while building the image.
- Helps create Docker images consistently and quickly.

DOCKERFILE INSTRUCTIONS

• FROM

Used to select the base image for the Docker image.

Example: FROM amazonlinux:2

Starts the Docker image using Amazon Linux.

• RUN

Used to execute commands while building the image.

Example: RUN yum install httpd -y

Installs Apache HTTP server inside the image.

• COPY

Used to copy files from the local system into the Docker image.

Example: COPY index.html /var/www/html/

Copies the HTML file into the Apache web folder.

• ADD

Used to add files or folders into the image.

It can also download files from URLs.

Example: ADD index.html /var/www/html/

• WORKDIR

Used to set the working directory inside the container.

Example: WORKDIR /var/www/html

Changes the current directory to /var/www/html.

- EXPOSE

Used to open a port for the container.

Example: EXPOSE 80

Opens port 80 for web access.

- CMD

Used to run the default command when the container starts.

Example: CMD ["/usr/sbin/httpd", "-D", "FOREGROUND"]

Starts the Apache server when the container runs.

PUSH IMAGE TO DOCKER HUB

Docker Tag Command

`docker tag <source-image>:<tag> <dockerhub-username>/<image-name>:<tag>`

Example

`docker tag myimage:latest jaisurya07/myimage:latest`

Used to create a Docker Hub compatible tag for the image.

Docker Push Command

`docker push <dockerhub-username>/<image-name>:<tag>`

Example

`docker push jaisurya07/myimage:latest`

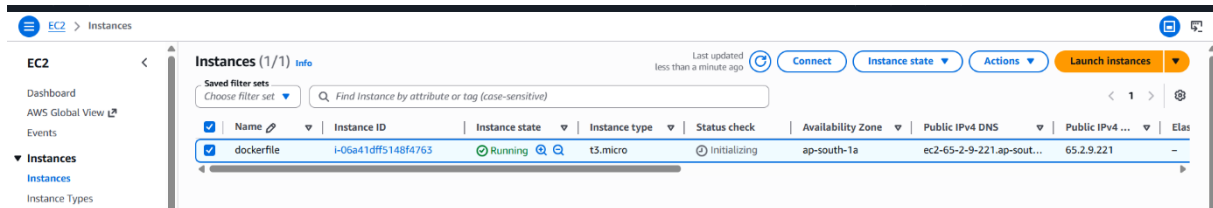
Used to upload the Docker image to the Docker Hub repository.

HANDS-ON

PUSH IMAGE TO DOCKER HUB

Screenshot: Create Linux EC2 Instance

- Launch a Linux EC2 instance in AWS
- Linux instance created successfully



Screenshot: Connect and Install Docker

- Connect to the Linux EC2 instance
- Execute `sudo su`
- Execute `yum update -y`
- Execute `yum install docker -y`
- Docker installed successfully



Screenshot: Start and Enable Docker Service

- Execute `systemctl start docker`
- Execute `systemctl enable docker`
- Execute `systemctl status docker`
- Docker service started and running successfully

```

[root@ip-172-31-34-209 ec2-user]# systemctl start docker
[root@ip-172-31-34-209 ec2-user]# systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[root@ip-172-31-34-209 ec2-user]# systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-05-26 01:17:04 UTC; 18s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 27656 (dockerd)
      Tasks: 9
     Memory: 31.7M
        CPU: 358ms
    CGroup: /system.slice/docker.service
            └─27656 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

May 26 01:17:03 ip-172-31-34-209.ap-south-1.compute.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
May 26 01:17:03 ip-172-31-34-209.ap-south-1.compute.internal docker[27656]: time="2026-05-26T01:17:03.264132287Z" level=info msg="Starting Docker Engine"

```

Screenshot: Login to Docker Hub

- Execute docker login
- Enter Docker Hub username
- Enter Docker Hub password
- Docker Hub login completed successfully

```

[root@ip-172-31-34-209 ec2-user]#
[root@ip-172-31-34-209 ec2-user]# docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head over to https://hub.docker.com/ to create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and is required for organizations using SSO. Learn more at https://docs.docker.com/go/access-tokens/
Username: jaisurya07
Password:
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded

```

Screenshot: Pull HTTPD Docker Image

- Execute docker pull httpd
- Apache HTTPD Docker image downloaded successfully from Docker Hub

```

[root@ip-172-31-34-209 ec2-user]#
[root@ip-172-31-34-209 ec2-user]# docker pull httpd
Using default tag: latest
latest: Pulling from library/httpd
5b4d6ff92fc4: Pull complete
f086d7c0b862: Pull complete
4f4fb700ef54: Pull complete
f60b6f23733a: Pull complete

```

Screenshot: Verify Downloaded HTTPD Image

- Execute docker images
- Downloaded httpd Docker image displayed successfully

```

[root@ip-172-31-34-209 ec2-user]#
[root@ip-172-31-34-209 ec2-user]# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
httpd          latest   b9097c67d4a2   6 days ago    117MB

```

Screenshot: Tag Docker Image

Docker Tag Syntax

docker tag <source-image>:<tag> <dockerhub-username>/<new-image-name>:<tag>

- Execute docker tag httpd:latest jaisurya07/httpd:latest
- HTTPD Docker image tagged successfully for Docker Hub push

```
[root@ip-172-31-34-209 ec2-user]#
[root@ip-172-31-34-209 ec2-user]# docker tag httpd:latest jaisurya07/httpd:latest
[root@ip-172-31-34-209 ec2-user]#
```

Screenshot: Push Docker Image to Docker Hub

Docker Push Syntax

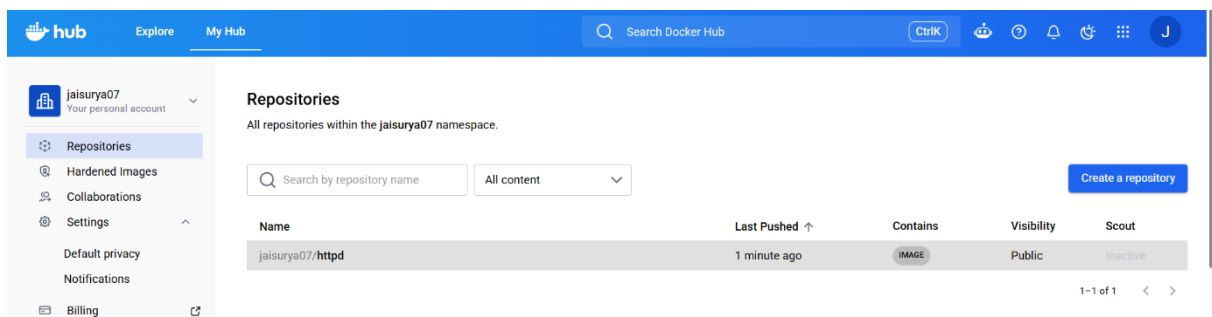
docker push <dockerhub-username>/<image-name>:<tag>

- Execute docker push jaisurya07/httpd:latest
- Docker image uploaded successfully to Docker Hub repository

```
[root@ip-172-31-34-209 ec2-user]# docker push jaisurya07/httpd:latest
The push refers to repository [docker.io/jaisurya07/httpd]
249013484d13: Pushed
845aa03d4e25: Pushed
911ed1d6053c: Pushed
5f70bf18a086: Pushed
7415a0d58160: Pushed
219a998c6050: Pushed
latest: digest: sha256:4393eb142d574ef8691fca90b9204203bdf6ba282150eff1009a23eb5bd44e2 size: 1572
[root@ip-172-31-34-209 ec2-user]#
```

Screenshot: Verify Docker Hub Repository

- Open Docker Hub website
- Login to Docker Hub account
- Open Repositories section
- Uploaded httpd image displayed successfully in the repository



Screenshot: Push Nginx Docker Image to Docker Hub

- Execute docker pull nginx
- Nginx Docker image downloaded successfully
- Execute docker tag nginx:latest jaisurya07/nginx:latest
- Nginx Docker image tagged successfully for Docker Hub
- Execute docker push jaisurya07/nginx:latest
- Nginx Docker image uploaded successfully to Docker Hub repository

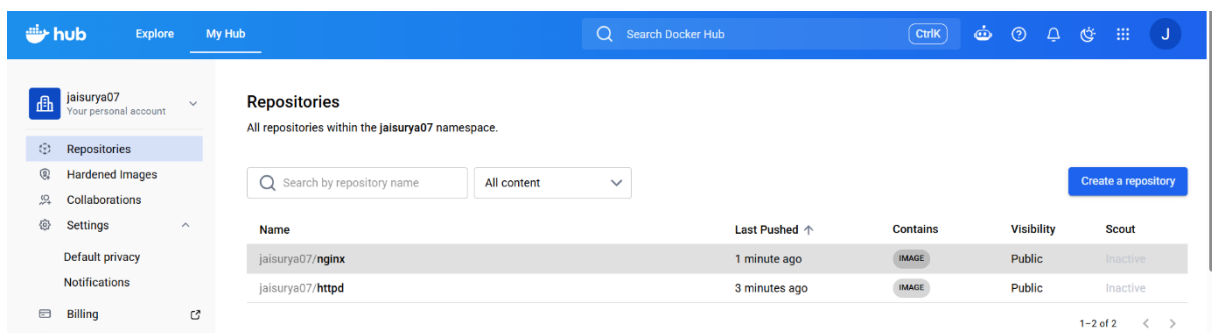
```

17 docker pull nginx
18 docker tag nginx:latest jaisurya07/nginx:latest
19 docker push jaisurya07/nginx:latest
20 history

```

Screenshot: Verify Docker Hub Repository

- Open Docker Hub website
- Login to Docker Hub account
- Open Repositories section
- Uploaded httpd and nginx Docker images displayed successfully in the repository



```

[root@ip-172-31-34-209 ec2-user]# docker images
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
jaisurya07/nginx    latest     7aaca76c508f  3 days ago    161MB
nginx               latest     7aaca76c508f  3 days ago    161MB
jaisurya07/httpd    latest     b9097c67d4a2  6 days ago    117MB
jaisurya07/httpd    latest1    b9097c67d4a2  6 days ago    117MB
httpd               latest     b9097c67d4a2  6 days ago    117MB
[root@ip-172-31-34-209 ec2-user]#

```

DOCKERFILE TO IMAGE

Screenshot: Create Dockerfile

- Execute vi Dockerfile
- Dockerfile opened successfully in the Linux instance editor

```
[root@ip-172-31-34-209 ec2-user]#  
[root@ip-172-31-34-209 ec2-user]# vi Dockerfile
```

Screenshot: Add Content to Dockerfile

- Press i to enter insert mode in vi editor
- Paste the Dockerfile configuration code
- Press Esc
- Type :wq! and press Enter
- Dockerfile saved successfully

```
FROM amazonlinux:2  
  
# Install Apache  
RUN yum update -y && \  
    yum install -y httpd && \  
    yum clean all  
  
# Apache web root  
WORKDIR /var/www/html  
  
# Copy your website files  
COPY . .  
  
# Expose Apache port  
EXPOSE 80  
  
# Start Apache in foreground  
CMD ["/usr/sbin/httpd", "-D", "FOREGROUND"]  
  
-- INSERT --
```

i-06a41dff5148f4763 (dockerfile)
PublicPc: 65.2.9.221 PrivatePc: 172.31.34.209

Screenshot: Create HTML File

- Execute vi index.html
- HTML file opened successfully in the vi editor

```
[root@ip-172-31-34-209 ec2-user]#  
[root@ip-172-31-34-209 ec2-user]# vi index.html
```

Screenshot: Add Content to HTML File

- Press i to enter insert mode in vi editor
- Paste the HTML code into index.html
- Press Esc

- Type :wq! and press Enter
- HTML file saved successfully

```
<!DOCTYPE html>
<html>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

Screenshot: Verify Dockerfile Content

- Execute cat Dockerfile
- Dockerfile content displayed successfully in the terminal

```
[root@ip-172-31-34-209 ec2-user]# cat Dockerfile
FROM amazonlinux:2

# Install Apache
RUN yum update -y && \
    yum install -y httpd && \
    yum clean all

# Apache web root
WORKDIR /var/www/html

# Copy your website files
COPY . .

# Expose Apache port
EXPOSE 80

# Start Apache in foreground
CMD ["/usr/sbin/httpd", "-D", "FOREGROUND"]
```

Screenshot: Build Docker Image from Dockerfile

- Execute docker build -t myimage .
- Docker image built successfully using the Dockerfile
- Image tagged with the name myimage

```
[root@ip-172-31-34-209 ec2-user]# docker build -t myimage .
[+] Building 21.8s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 390B
=> [internal] load metadata for docker.io/library/amazonlinux:2
=> [auth] library/amazonlinux:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/amazonlinux:2@sha256:22d5bfff62c7d84c3f7b39e2697fd04a2262c1c97d55a028f4a8f233493b906eb
=> => resolve docker.io/library/amazonlinux:2@sha256:22d5bfff62c7d84c3f7b39e2697fd04a2262c1c97d55a028f4a8f233493b906eb
=> => sha256:c741d741503511d961c454b4f2ea18650ce53cf7194f8alf8924c6c674404555 1.02kB / 1.02kB
=> => sha256:91f54f52a602fc7e10226e8bbace2d9ac6ba7940e510177b177ea773485a2a52 586B / 586B
=> => sha256:45dd431b24e3561020a83f5e29dd2642d9ec2f2788c6826e5b1e9ee25ee38759 62.95MB / 62.95MB
=> => sha256:22d5bfff62c7d84c3f7b39e2697fd04a2262c1c97d55a028f4a8f233493b906eb 2.37kB / 2.37kB
=> => extracting sha256:45dd431b24e3561020a83f5e29dd2642d9ec2f2788c6826e5b1e9ee25ee38759
=> [internal] load build context
=> => transferring context: 2.17kB
=> [2/4] RUN yum update -y && yum install -y httpd && yum clean all
=> [3/4] WORKDIR /var/www/html
```

Screenshot: Verify Built Docker Image

- Execute docker images
- Newly created myimage Docker image displayed successfully

```
[root@ip-172-31-34-209 ec2-user]# docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
myimage              latest          2fd9dee6a7d0   About a minute ago  246MB
jaisurya07/nginx     latest          7aaca76c508f   3 days ago      161MB
nginx                latest          7aaca76c508f   3 days ago      161MB
jaisurya07/httpd     latest          b9097c67d4a2   6 days ago      117MB
jaisurya07/httpd     latest1         b9097c67d4a2   6 days ago      117MB
httpd                latest          b9097c67d4a2   6 days ago      117MB
[root@ip-172-31-34-209 ec2-user]#
```

Screenshot: Run Docker Container from Custom Image

- Execute docker run -itd --name con1 -p 8080:80 myimage:latest
- Docker container con1 created and running successfully using myimage

```
[root@ip-172-31-34-209 ec2-user]#
[root@ip-172-31-34-209 ec2-user]# docker run -itd --name con1 -p 8080:80 myimage:latest
8b0bf6a1ee43d78e7e58be1a9a7c9ce6705d85338c26b95a57edf6180bdfd6d8
[root@ip-172-31-34-209 ec2-user]#
```

Screenshot: Access Custom Docker Web Page

- Copy the public IP address of the EC2 instance
- Open a new browser tab and enter Public-IP:8080
- Custom HTML web page content displayed successfully from the Docker container



My First Heading

My first paragraph.

Screenshot: Create Dockerfile Using ADD Instruction

- Execute vi Dockerfile
- Add Dockerfile configuration using the ADD instruction
- Save the Dockerfile successfully
- Execute vi index.html
- Add HTML content and save the file
- Execute docker build -t myimage .
- Docker image built successfully using ADD instruction

- Execute `docker run -itd --name con4 -p 8083:80 myimage:latest`
- Docker container created and running successfully using the updated image

```

28 vi Dockerfile
29 vi index.html
30 docker build -t myimage .
31 docker run -itd --name con2 -p 8081:80 myimage:latest
32 history
[con2@ip-172-31-34-200 ~]$

FROM amazonlinux:2

# Install httpd, unzip, wget
RUN yum update -y && \
    yum install -y httpd unzip && \
    yum clean all

# Download template zip using ADD
ADD https://templatemo.com/download/templatemo_606_string_master /tmp/template.zip

# Unzip template and move content to Apache root
RUN unzip /tmp/template.zip -d /tmp && \
    cp -r /tmp/templatemo_606_string_master/* /var/www/html/ && \
    rm -rf /tmp/template.zip /tmp/templatemo_606_string_master

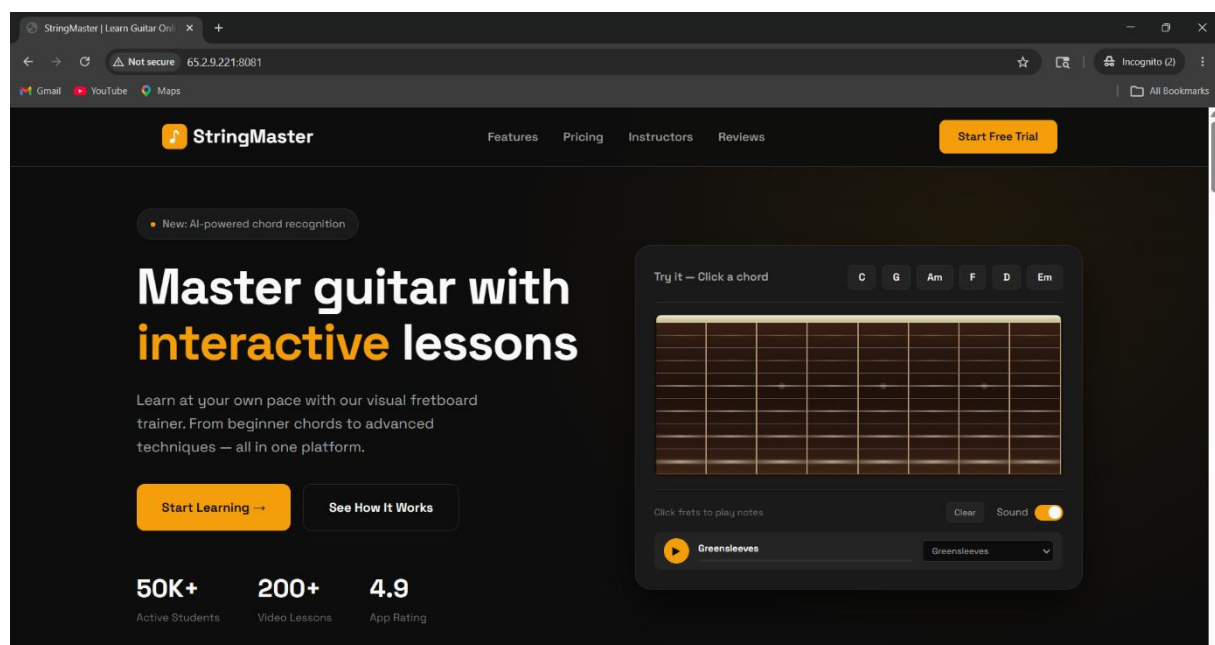
# Expose Apache port
EXPOSE 80

# Run Apache in foreground
CMD ["/usr/sbin/httpd", "-D", "FOREGROUND"]

```

Screenshot: Access Docker Container Web Page

- Copy the public IP address of the EC2 instance
- Open browser and enter Public-IP:8083
- Web page loaded successfully from the Docker container using ADD instruction



```
34 docker images
35 docker tag myimage:latest jaisurya07/myimage:latest
36 docker tag push jaisurya07/myimage:latest
37 docker push jaisurya07/myimage:latest
```

Screenshot: Tag and Push Custom Docker Image

- Execute `docker tag myimage:latest jaisurya07/myimage:latest`
- Docker image tagged successfully for Docker Hub
- Execute `docker push jaisurya07/myimage:latest`
- Custom Docker image uploaded successfully to Docker Hub repository

